PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop
for Autonomous Cognitive Systems

Drawing Sheets: 8 figures

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed January 31, 2026

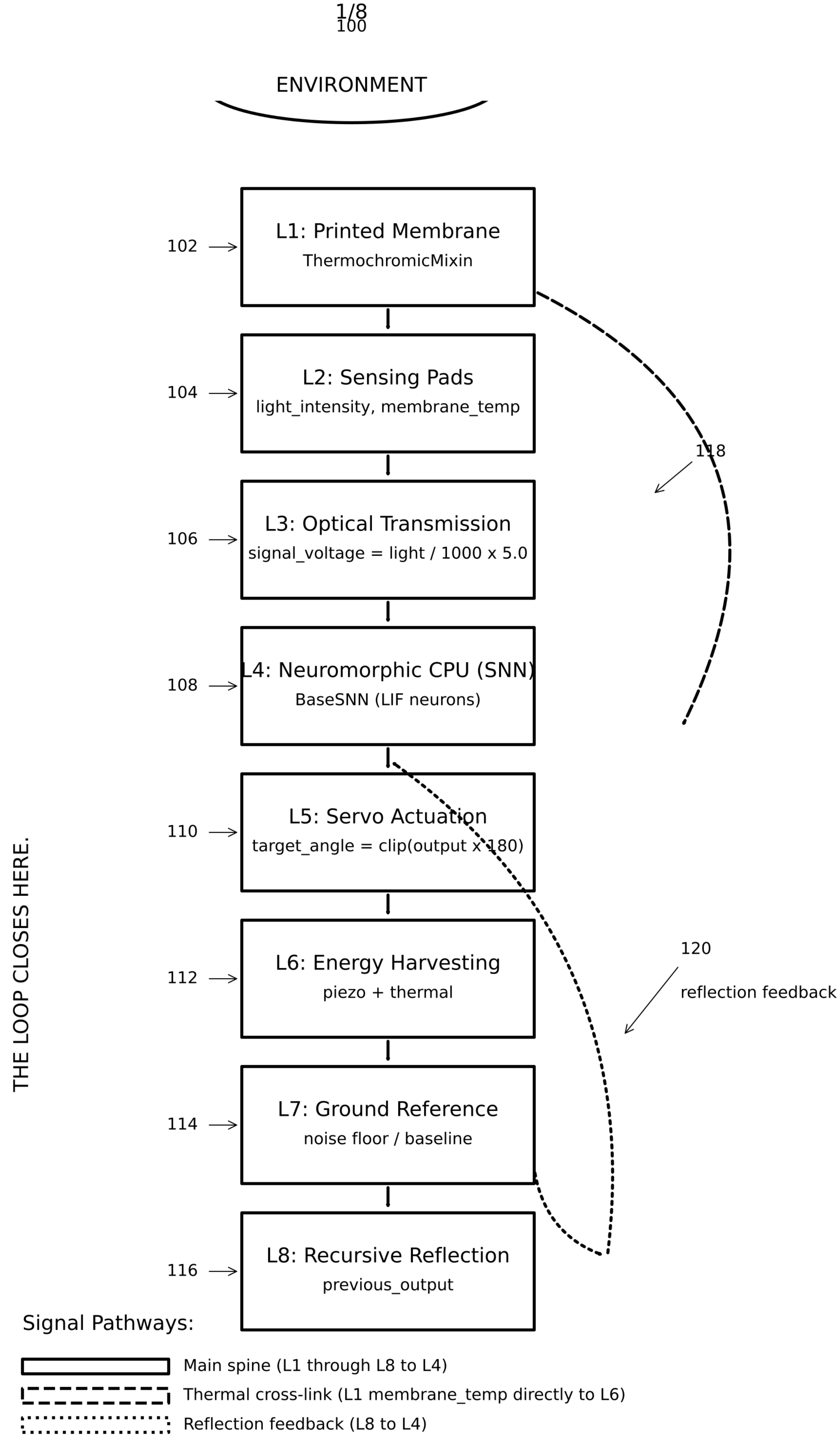
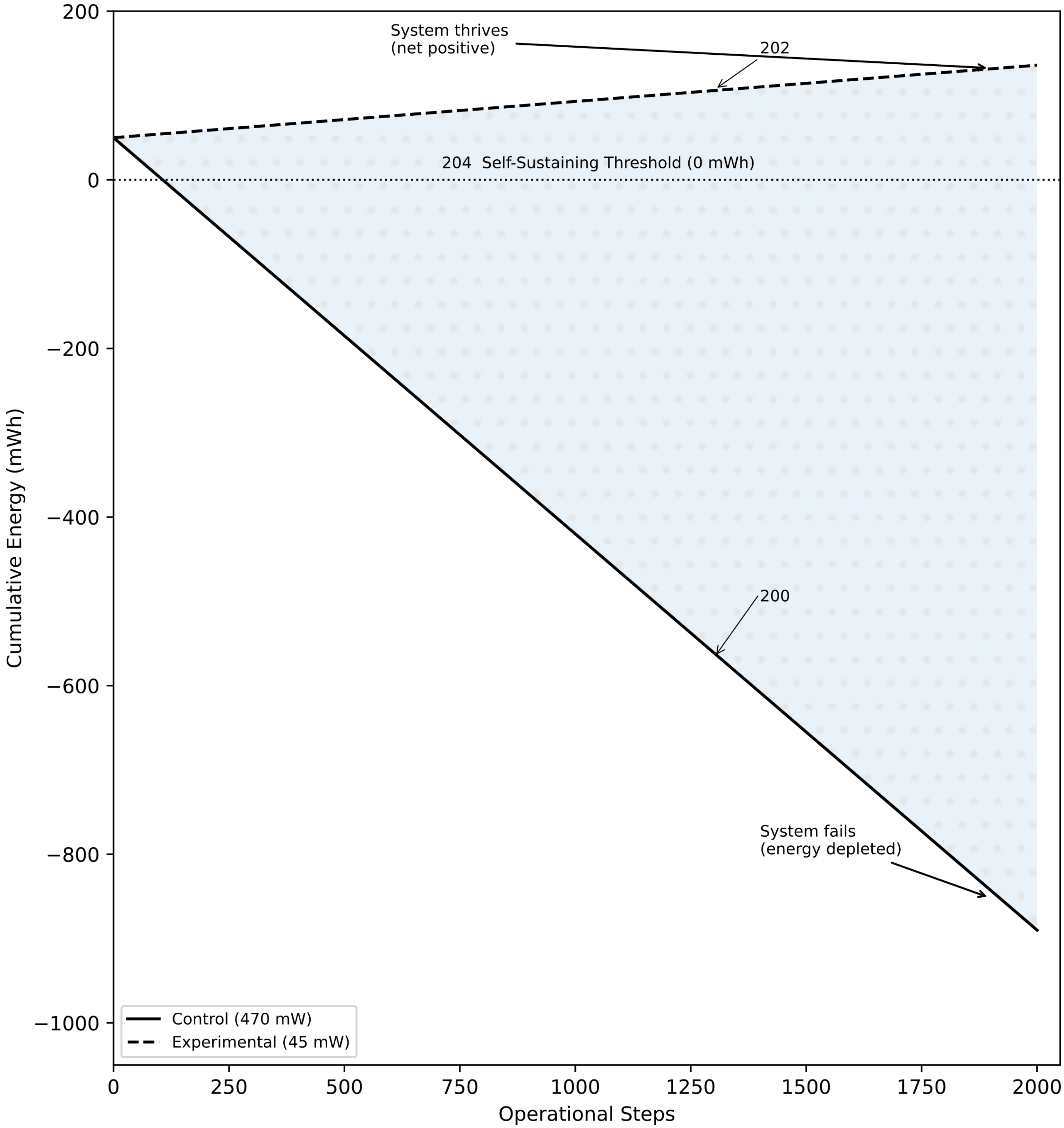

100
ENVIRONMENT
102
L1: Printed Membrane
ThermochromicMixin
104
L2: Sensing Pads
light_intensity, membrane_temp
106
L3: Optical Transmission
signal_voltage = light / 1000 x 5.0
108
L4: Neuromorphic CPU (SNN)
BaseSNN (LIF neurons)
110
L5: Servo Actuation
target_angle = clip(output x 180)
112
L6: Energy Harvesting
piezo + thermal
114
L7: Ground Reference
noise floor / baseline
116
L8: Recursive Reflection
previous_output
118
120
reflection feedback
THE LOOP CLOSES HERE.
Signal Pathways:
Main spine (L1 through L8 to L4)
Thermal cross-link (L1 membrane_temp directly to L6)
Reflection feedback (L8 to L4)

FIG. 1



System thrives
(net positive)
202
204 Self-Sustaining Threshold (0 mWh)
200
System fails
(energy depleted)
Control (470 mW)
Experimental (45 mW)
200
0
−200
−400
−600
−800
−1000
Cumulative Energy (mWh)
0
250
500
750
1000
1250
1500
1750
2000
Operational Steps

FIG. 2

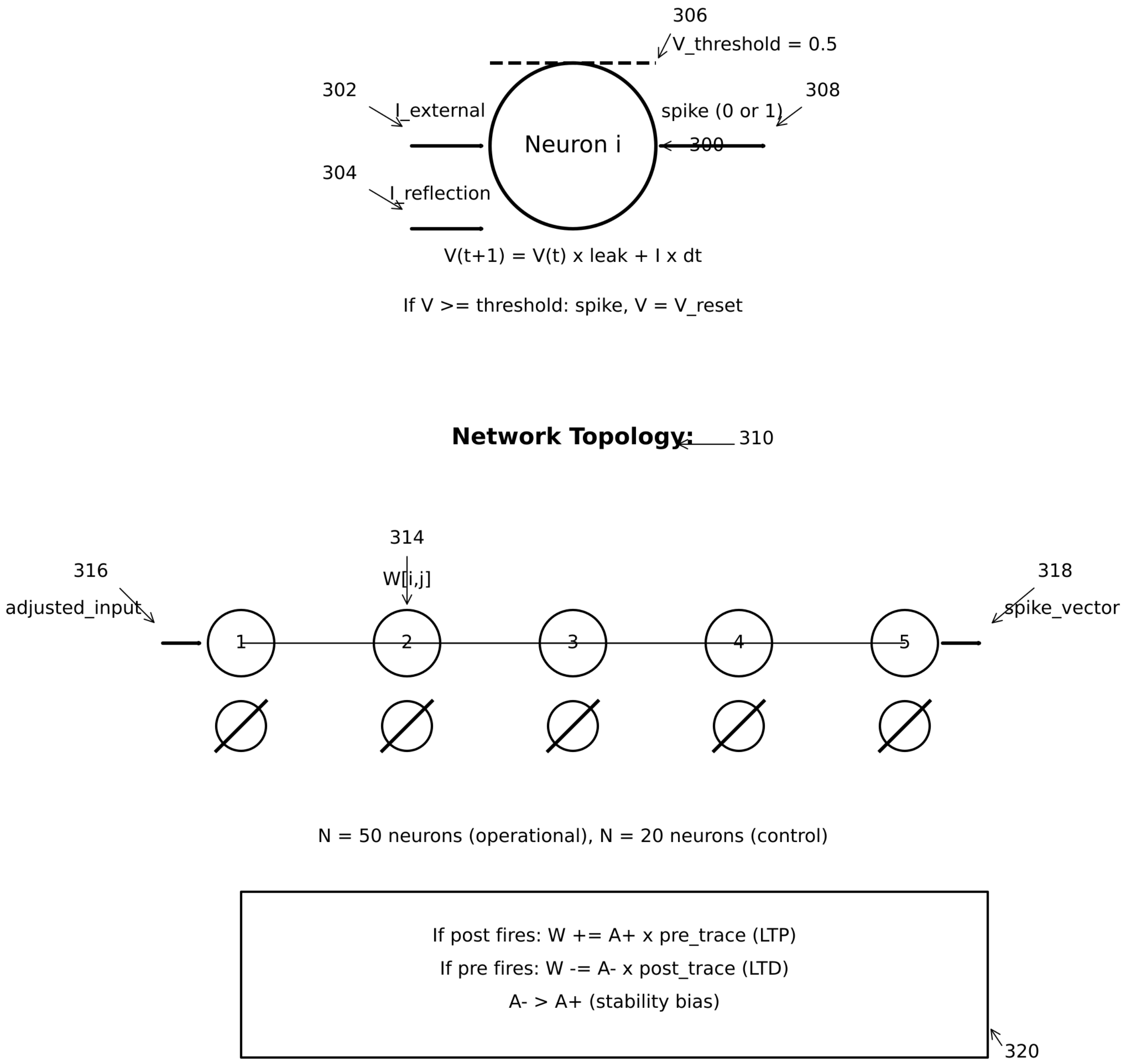

306
V_threshold = 0.5
302
I_external
Neuron i
spike (0 or 1)
308
300
304
I_reflection
V(t+1) = V(t) x leak + I x dt
If V >= threshold: spike, V = V_reset
Network Topology:
310
314
W[i,j]
316
adjusted_input
318
spike_vector
1
2
3
4
5
N = 50 neurons (operational), N = 20 neurons (control)
If post fires: W += A+ x pre_trace (LTP)
If pre fires: W -= A- x post_trace (LTD)
A- > A+ (stability bias)
320

FIG. 3

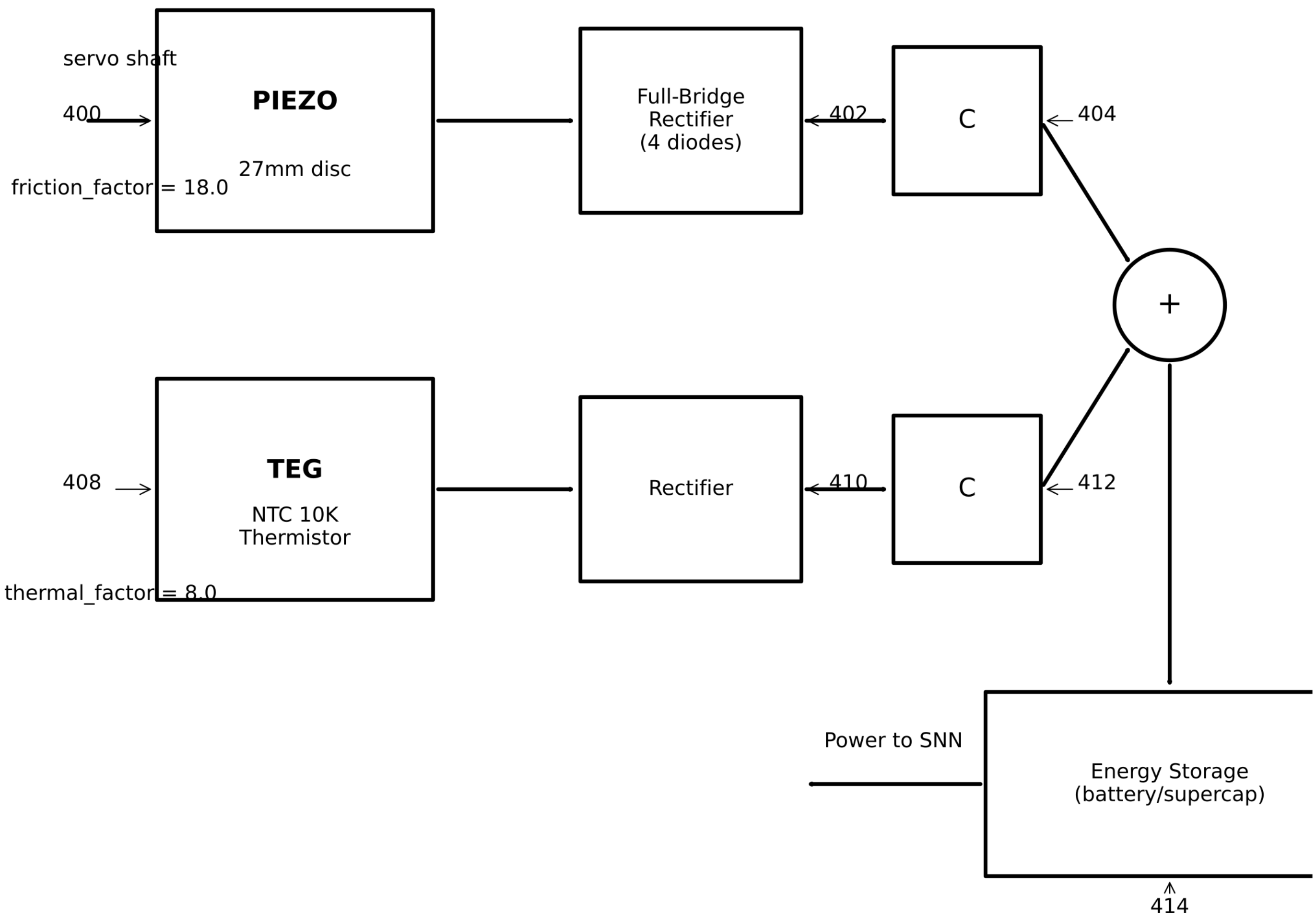

servo shaft
400
friction_factor = 18.0
PIEZO
27mm disc
Full-Bridge
Rectifier
(4 diodes)
402
C
404
+
408
thermal_factor = 8.0
TEG
NTC 10K
Thermistor
Rectifier
410
C
412
Power to SNN
Energy Storage
(battery/supercap)
414

FIG. 4

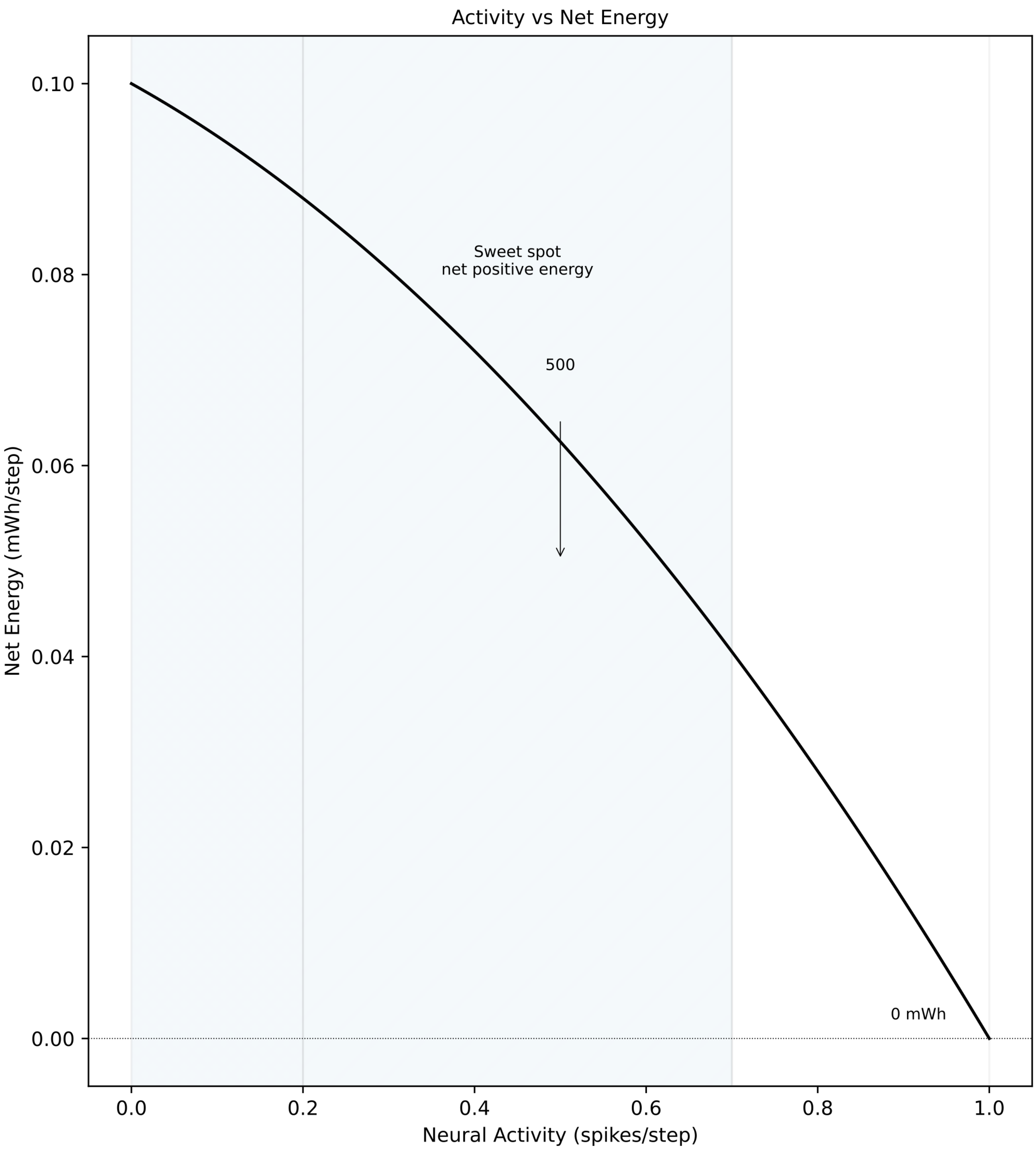
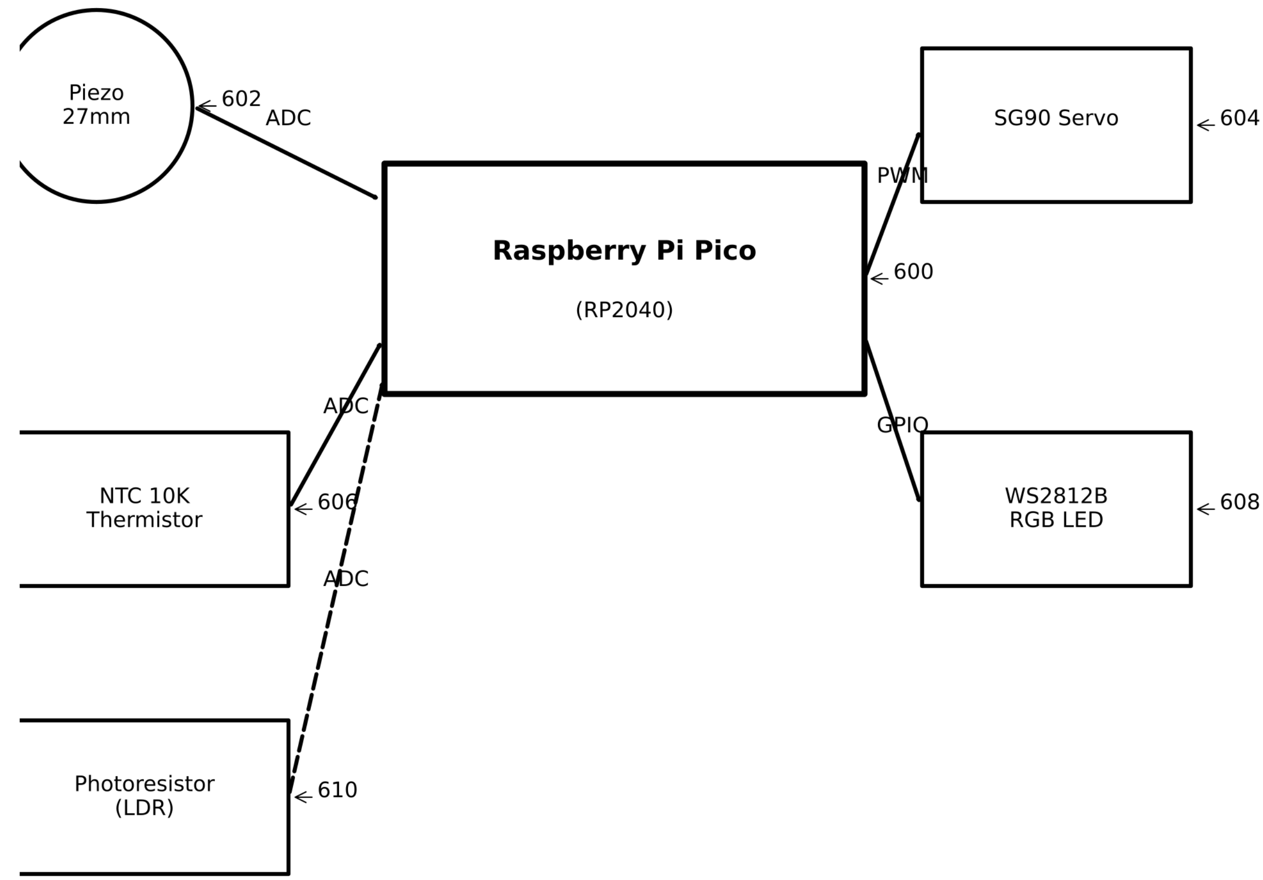


FIG. 5

Low activity
insufficient
harvesting

High activity
quadratic cost
dominates



Bill of Materials

Raspberry Pi Pico	$4
SG90 Micro Servo	$3
27mm Piezo Disc	$2
NTC 10K Thermistor	$1
Photoresistor (LDR)	$1
WS2812B RGB LED	$1
Misc (wires, board)	$3

Total BOM Cost: <$25

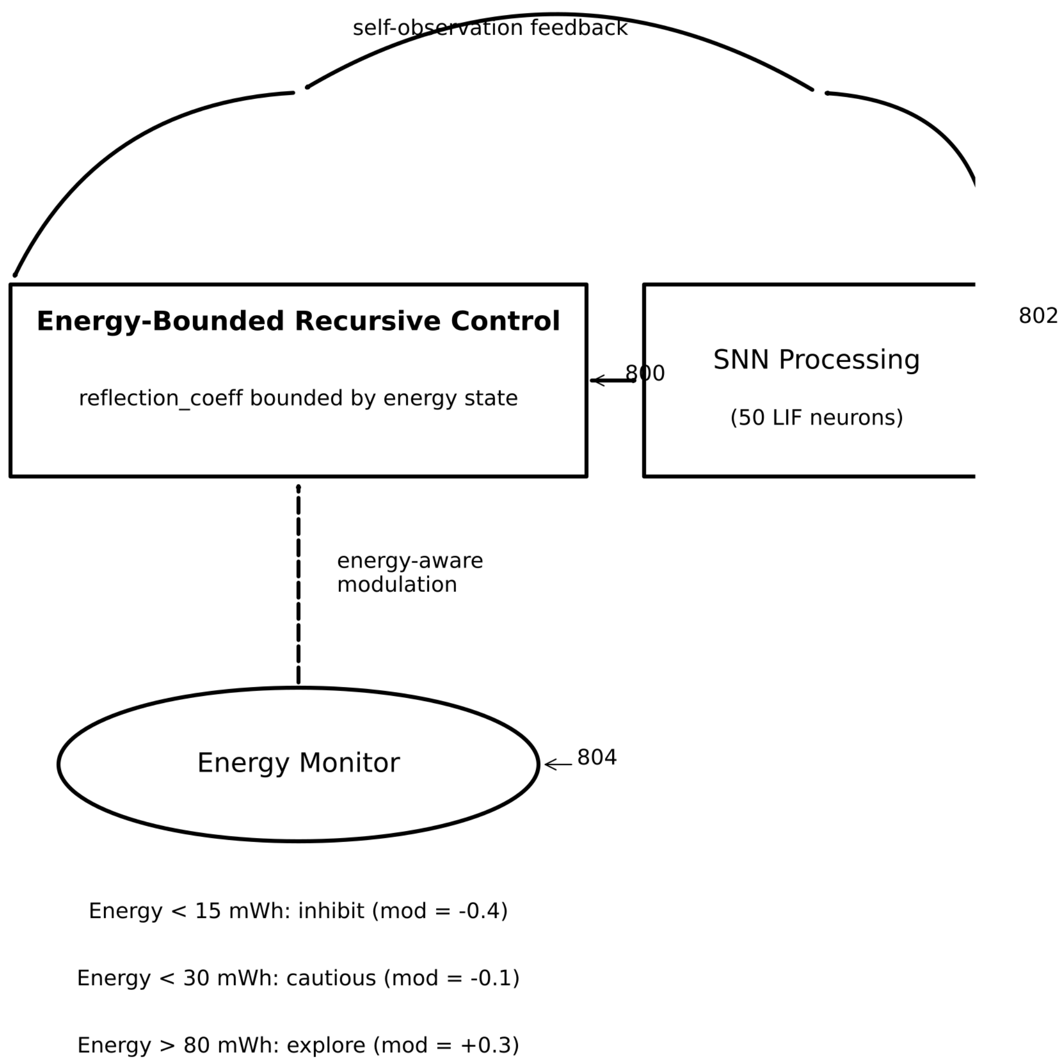
FIG. 6

Claim	Criterion	Measured	Threshold	Result
	Phase 7 Control Baseline (EnergyConfig)			
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	Std(θ)≤45°	≤45°	PASS
D	Input-output gain	corr≥0.2	≥0.2	PASS
E	Saturation ratio	sat≤0.20	≤0.20	PASS
	Phase 7 Full Integration (Control)			
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
	MCP Integration (BalancedEnergyConfig)			
--	Energy at 2000 steps	+4,170 mWh	> 0	CONFIRMED
	Phase 8 STDP			
F8.1	Weight entropy decrease	2.95→2.02	decrease	PASS
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	< 10%	PASS

700

All claims validated. System reproducible via git tags.

FIG. 7



Recursive control loop bounded by energy levels to ensure self-sustainability.

FIG. 8

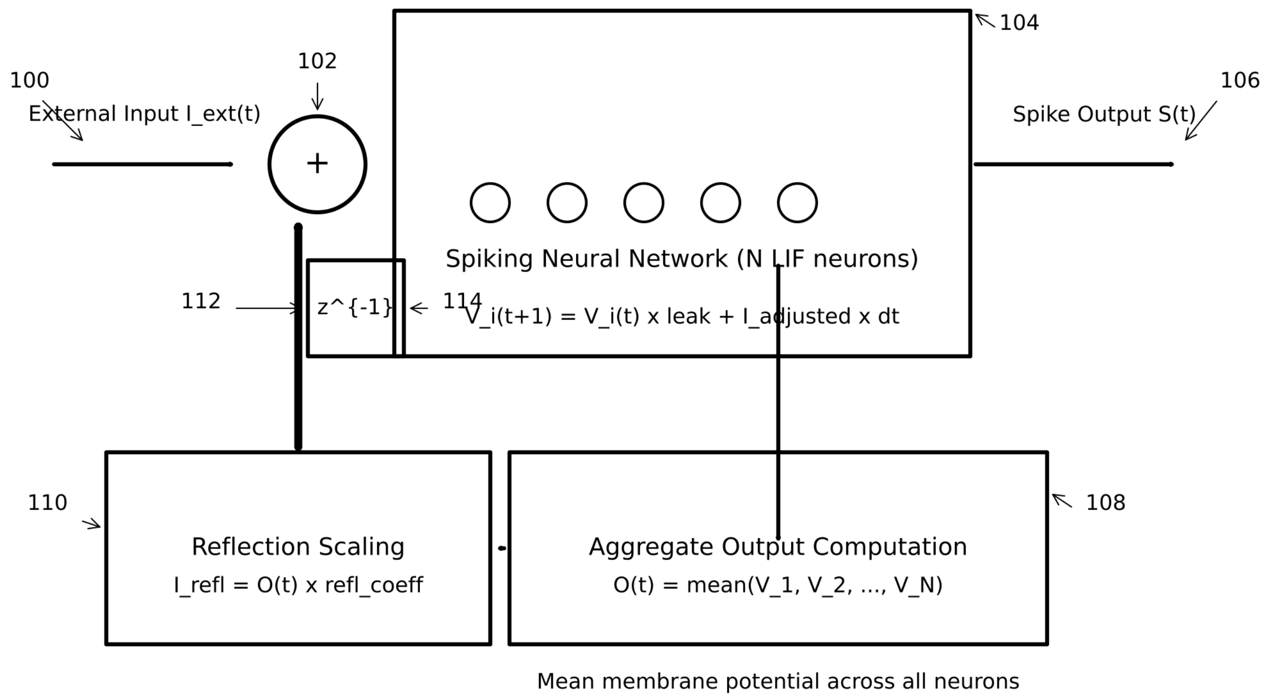
PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

Drawing Sheets: 6 figures

Inventor: Kevin Christopher Ward

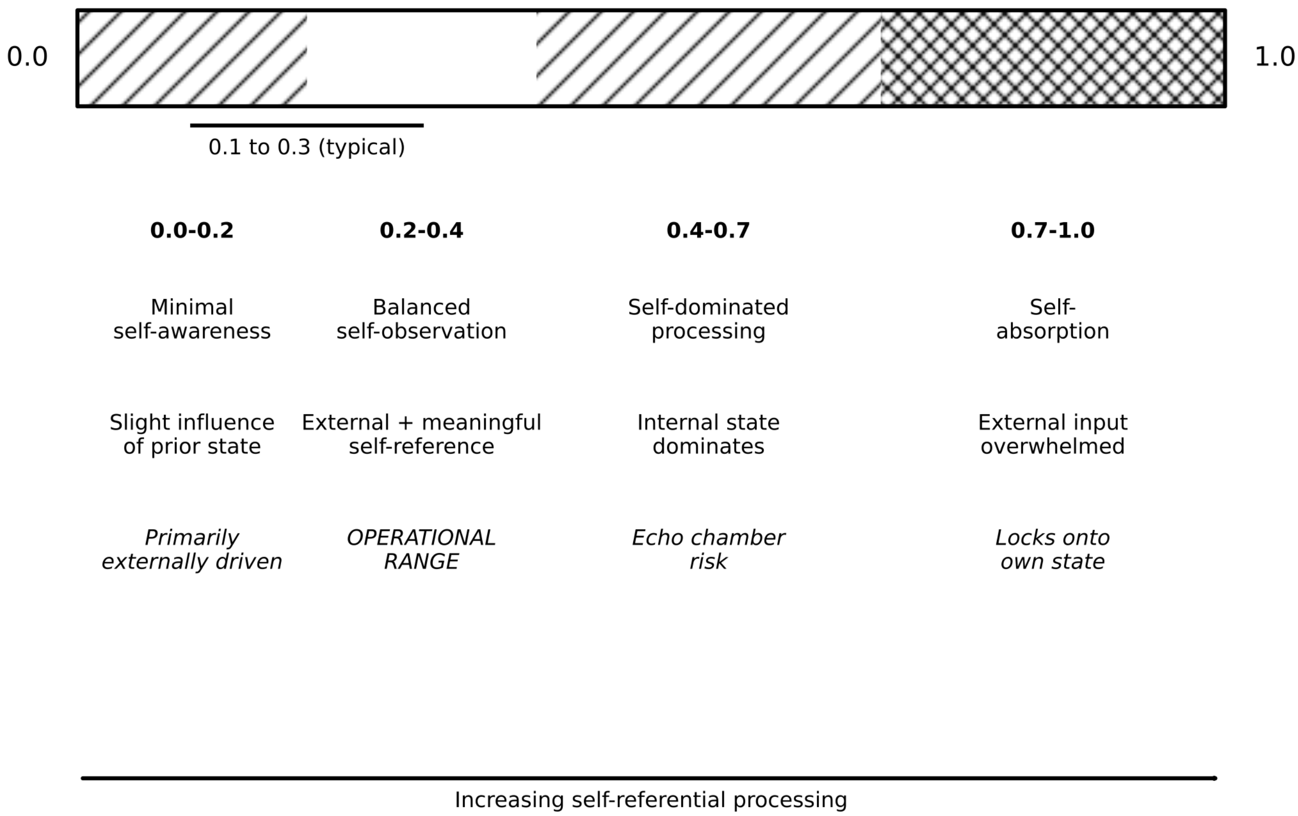
Provisional Patent Application — Filed January 31, 2026



Self-observation: the network observes its own prior aggregate state

I_adjusted(t+1) = I_ext(t+1) + I_reflection(t)

FIG. 1



The reflection coefficient is the system's self-awareness dial

FIG. 2

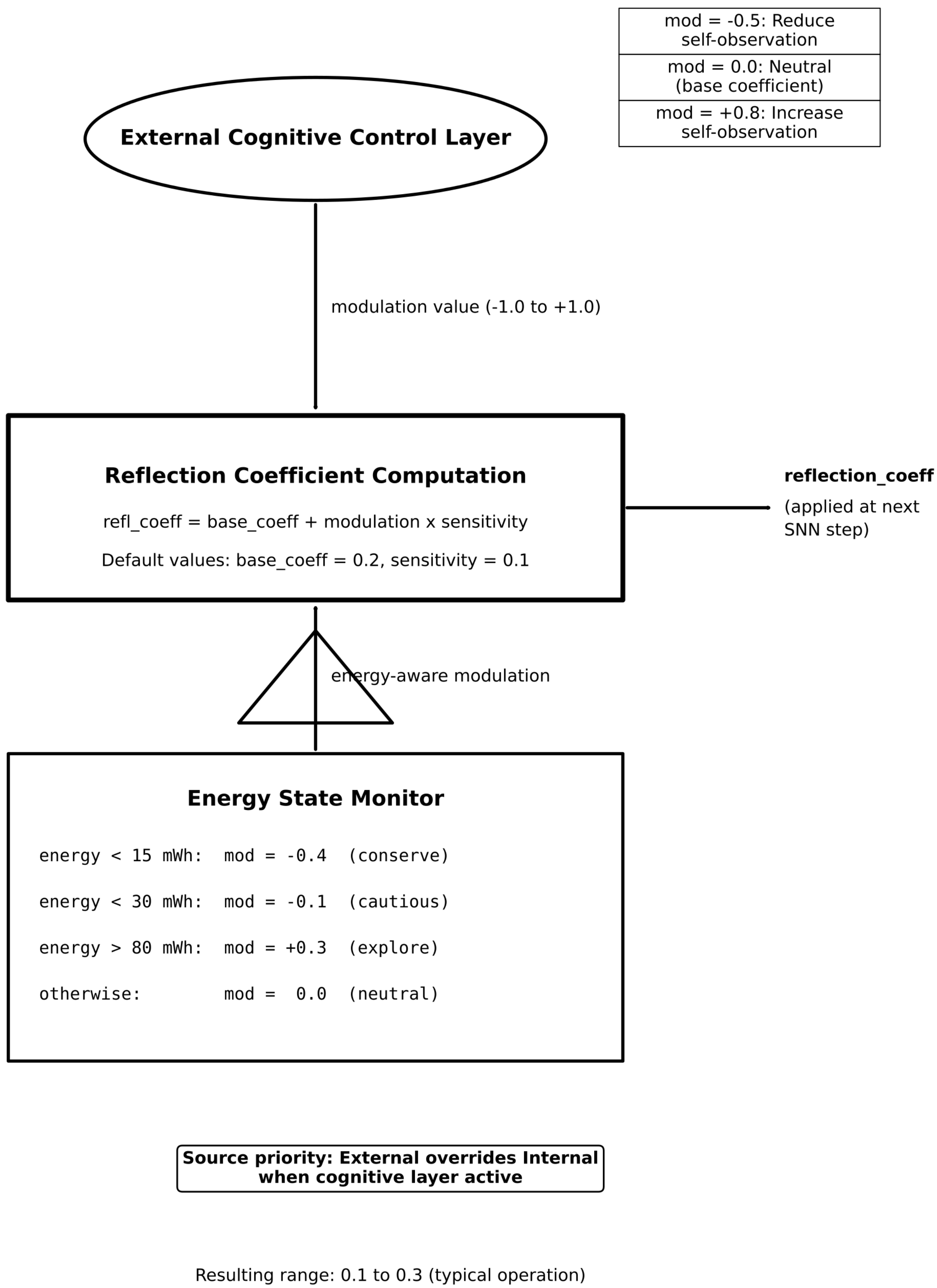

mod = -0.5: Reduce self-observation
mod = 0.0: Neutral (base coefficient)
mod = +0.8: Increase self-observation
External Cognitive Control Layer
modulation value (-1.0 to +1.0)
Reflection Coefficient Computation
refl_coeff = base_coeff + modulation x sensitivity
Default values: base_coeff = 0.2, sensitivity = 0.1
reflection_coeff
(applied at next SNN step)
energy-aware modulation
Energy State Monitor
energy < 15 mWh: mod = -0.4 (conserve)
energy < 30 mWh: mod = -0.1 (cautious)
energy > 80 mWh: mod = +0.3 (explore)
otherwise: mod = 0.0 (neutral)
Source priority: External overrides Internal when cognitive layer active
Resulting range: 0.1 to 0.3 (typical operation)

FIG. 3

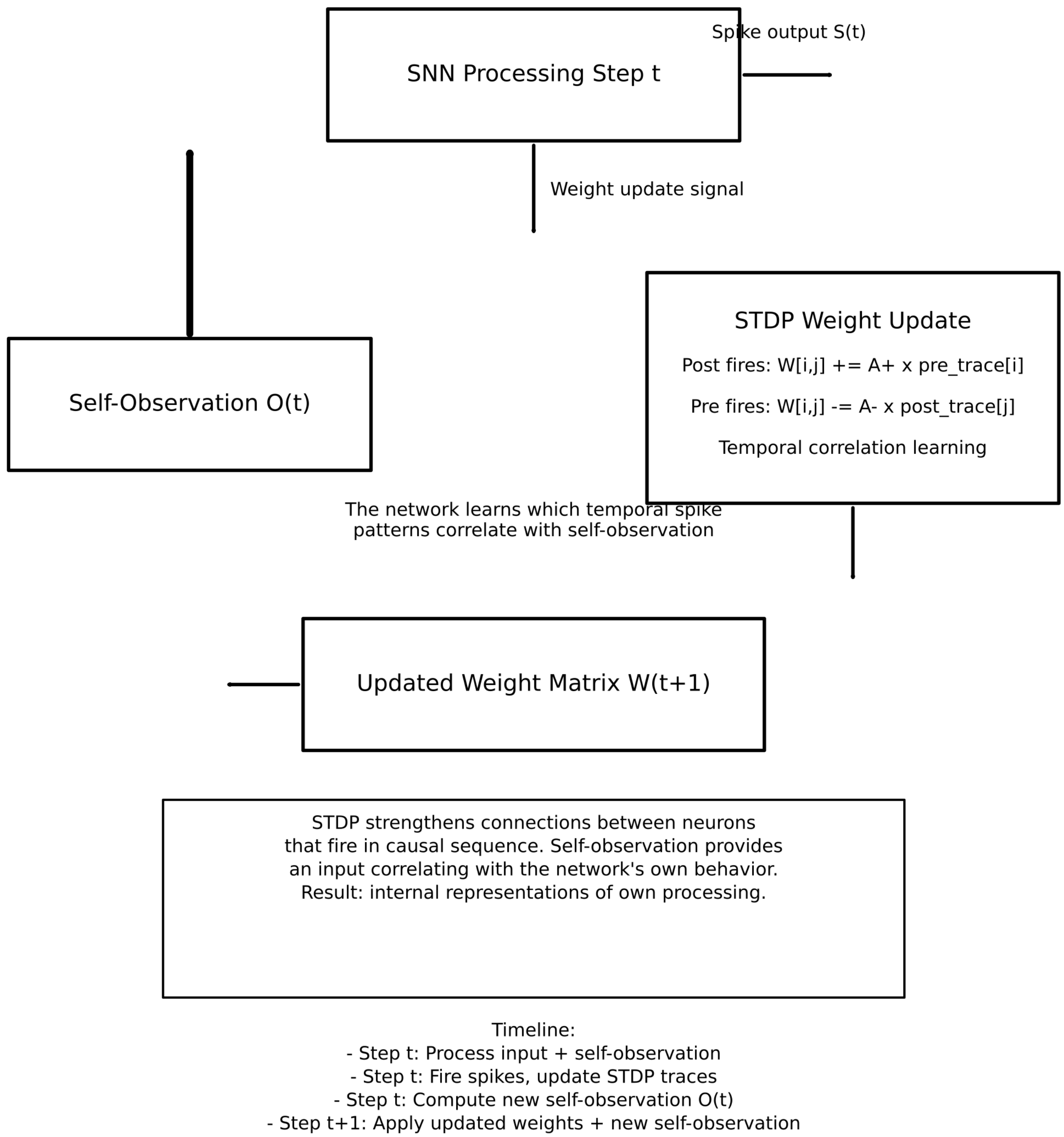
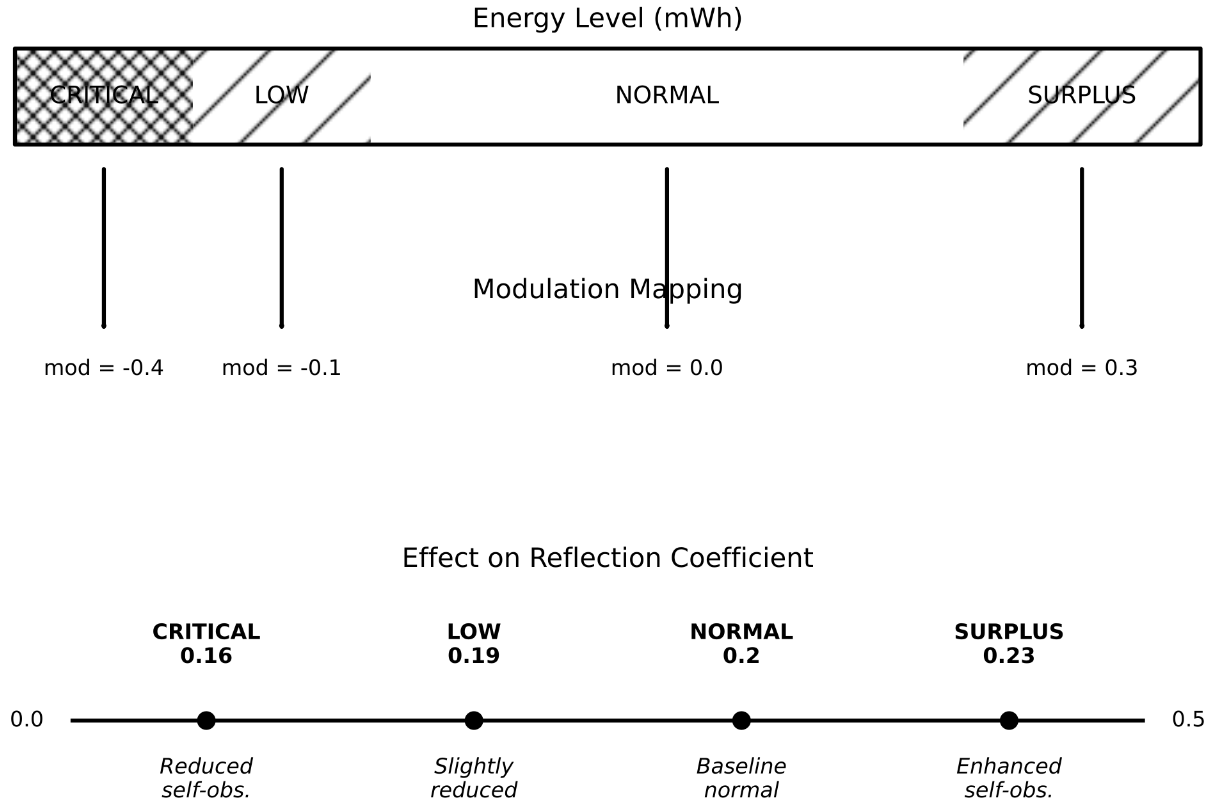

SNN Processing Step t
Spike output S(t)
Weight update signal
Self-Observation O(t)
STDP Weight Update
Post fires: W[i,j] += A+ x pre_trace[i]
Pre fires: W[i,j] -= A- x post_trace[j]
Temporal correlation learning
The network learns which temporal spike patterns correlate with self-observation
Updated Weight Matrix W(t+1)
STDP strengthens connections between neurons that fire in causal sequence. Self-observation provides an input correlating with the network's own behavior. Result: internal representations of own processing.
Timeline:
- Step t: Process input + self-observation
- Step t: Fire spikes, update STDP traces
- Step t: Compute new self-observation O(t)
- Step t+1: Apply updated weights + new self-observation

FIG. 4



Energy-aware regulation ensures the system reduces cognitive overhead when energy is scarce and explores deeper self-reference when energy is abundant

FIG. 5

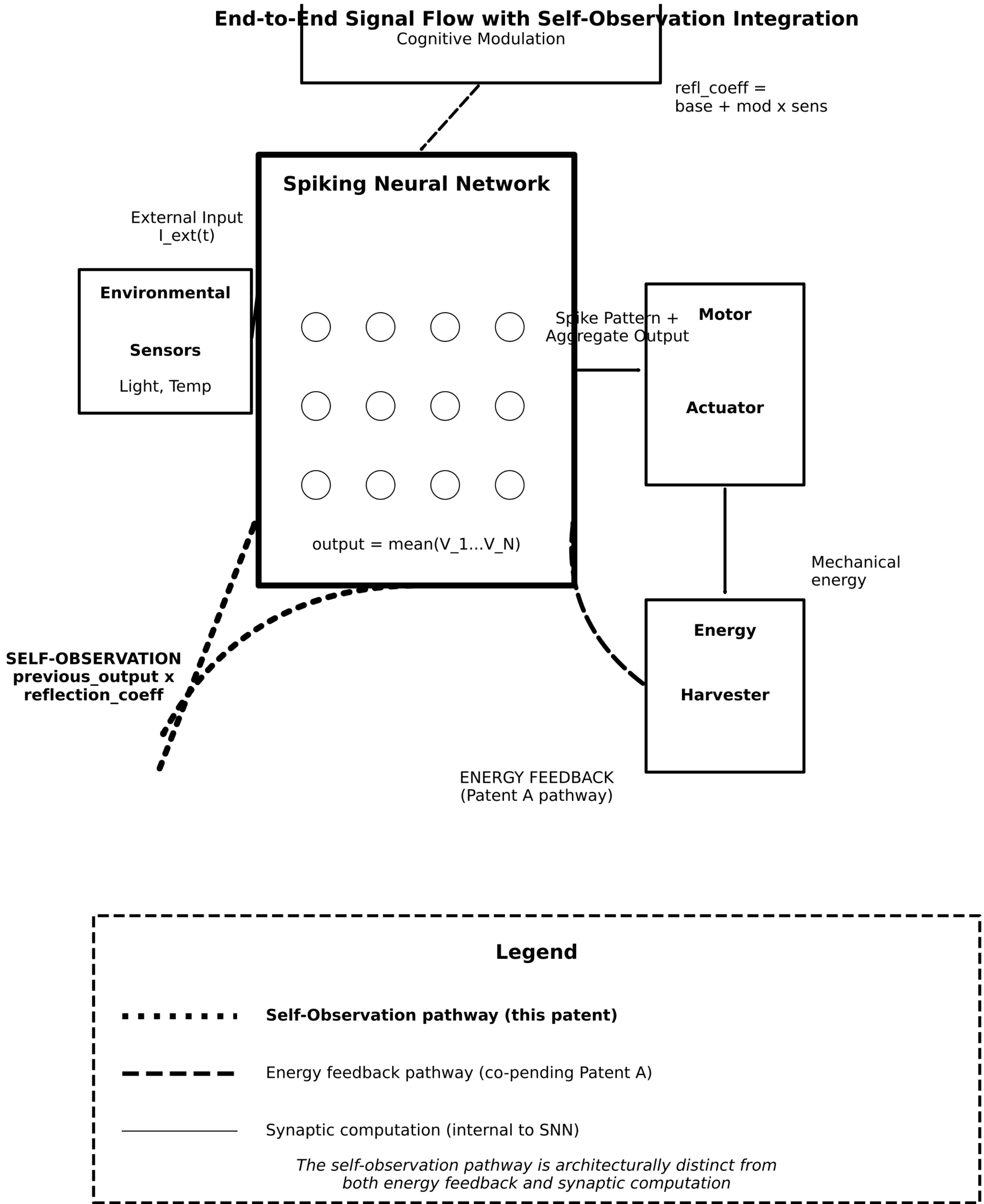

End-to-End Signal Flow with Self-Observation Integration
Cognitive Modulation
refl_coeff =
base + mod x sens
Spiking Neural Network
External Input
I_ext(t)
Environmental
Sensors
Light, Temp
Spike Pattern +
Aggregate Output
Motor
Actuator
output = mean(V_1...V_N)
Mechanical
energy
Energy
Harvester
SELF-OBSERVATION
previous_output x
reflection_coeff
ENERGY FEEDBACK
(Patent A pathway)
Legend
Self-Observation pathway (this patent)
Energy feedback pathway (co-pending Patent A)
Synaptic computation (internal to SNN)
The self-observation pathway is architecturally distinct from
both energy feedback and synaptic computation

FIG. 6

PATENT C

Method and System for Autonomous Self-Regulation
and Cognitive Resynchronization in Neural Processing
Systems During Disconnection from External Control Layers

Drawing Sheets: 7 figures

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed January 31, 2026

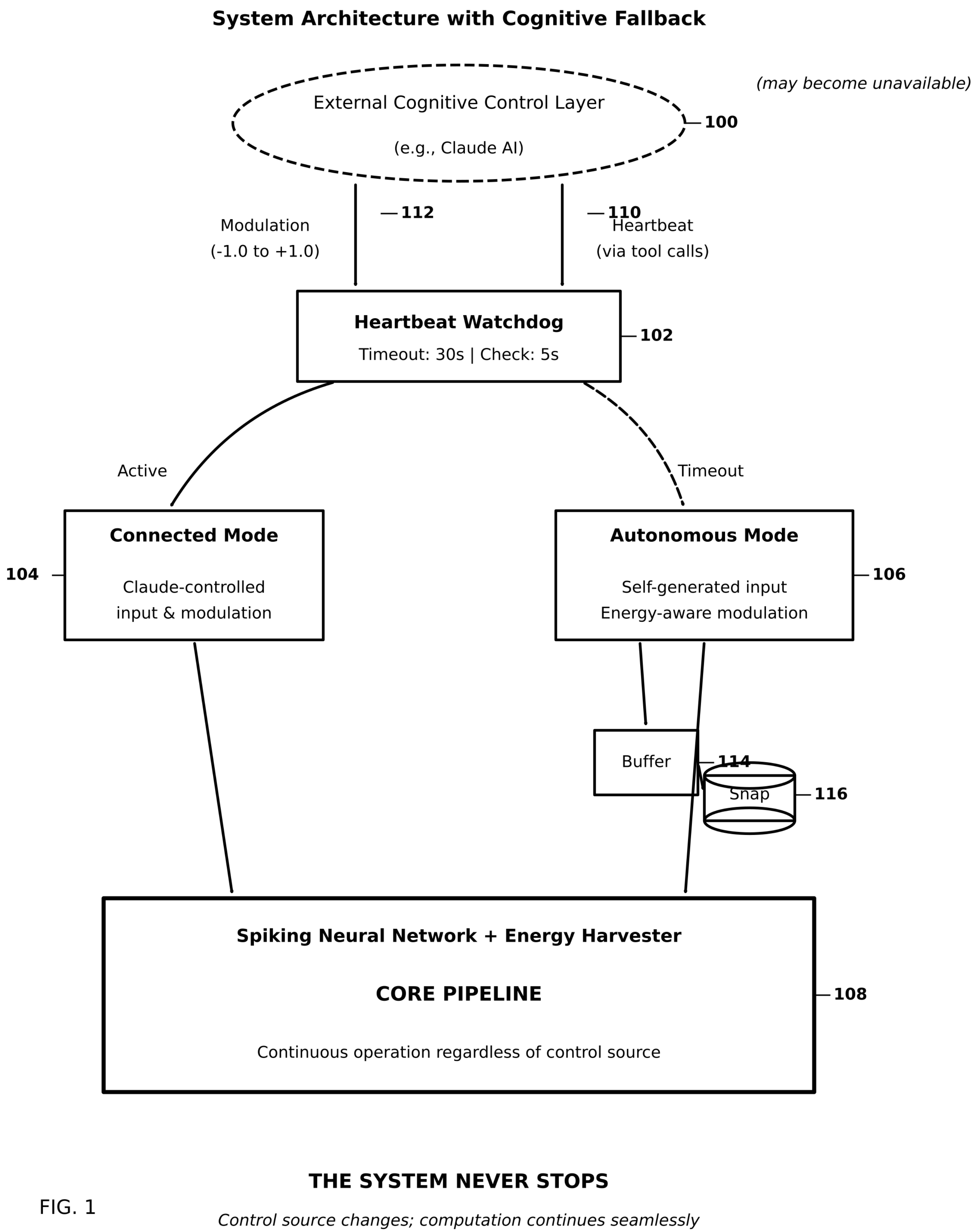

System Architecture with Cognitive Fallback
(may become unavailable)
External Cognitive Control Layer
(e.g., Claude AI)
100
Modulation
(-1.0 to +1.0)
112
110
Heartbeat
(via tool calls)
Heartbeat Watchdog
Timeout: 30s | Check: 5s
102
Active
Timeout
Connected Mode
Claude-controlled
input & modulation
104
Autonomous Mode
Self-generated input
Energy-aware modulation
106
Buffer
114
Snap
116
Spiking Neural Network + Energy Harvester
CORE PIPELINE
Continuous operation regardless of control source
108
THE SYSTEM NEVER STOPS
Control source changes; computation continues seamlessly

FIG. 1

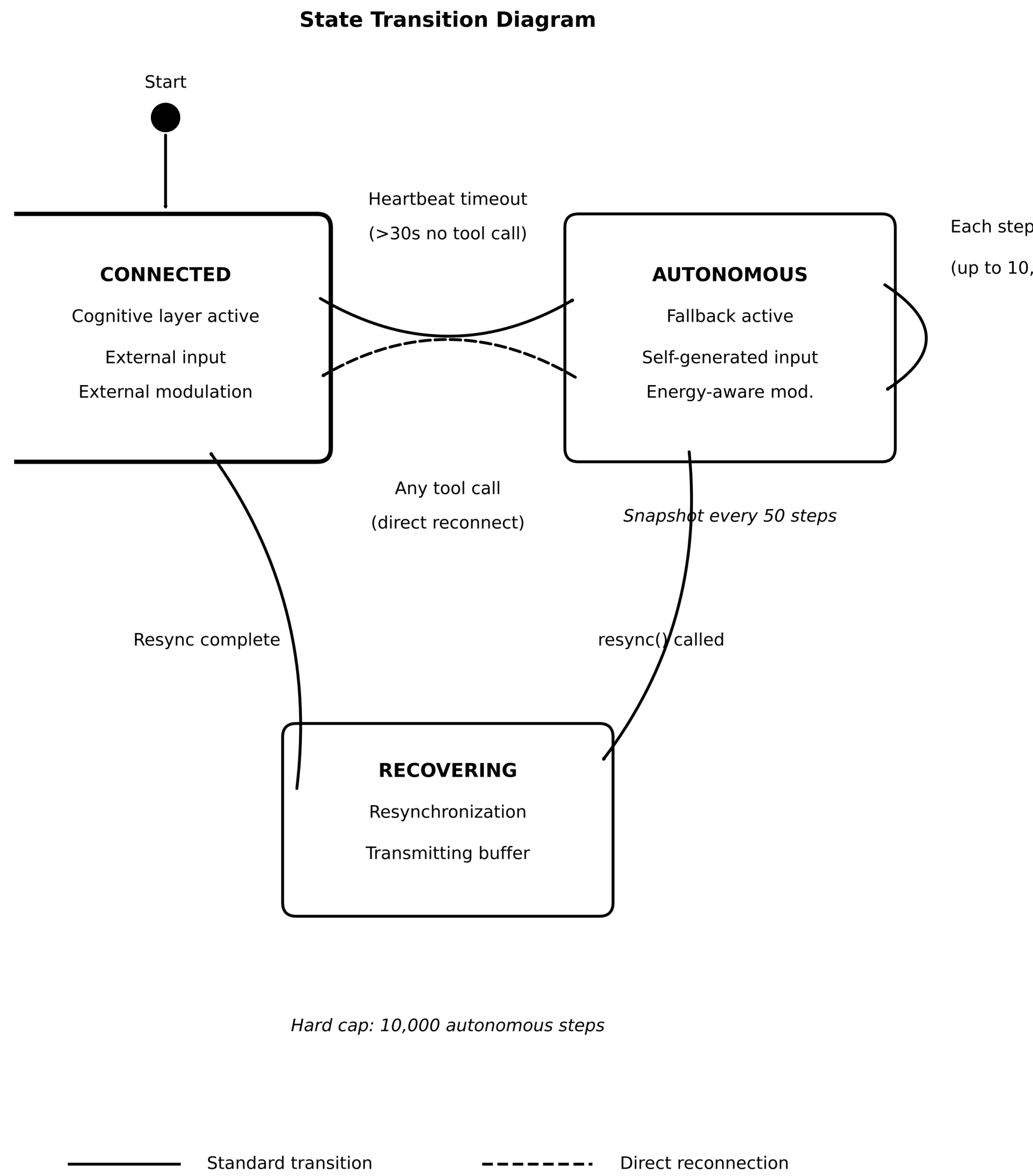
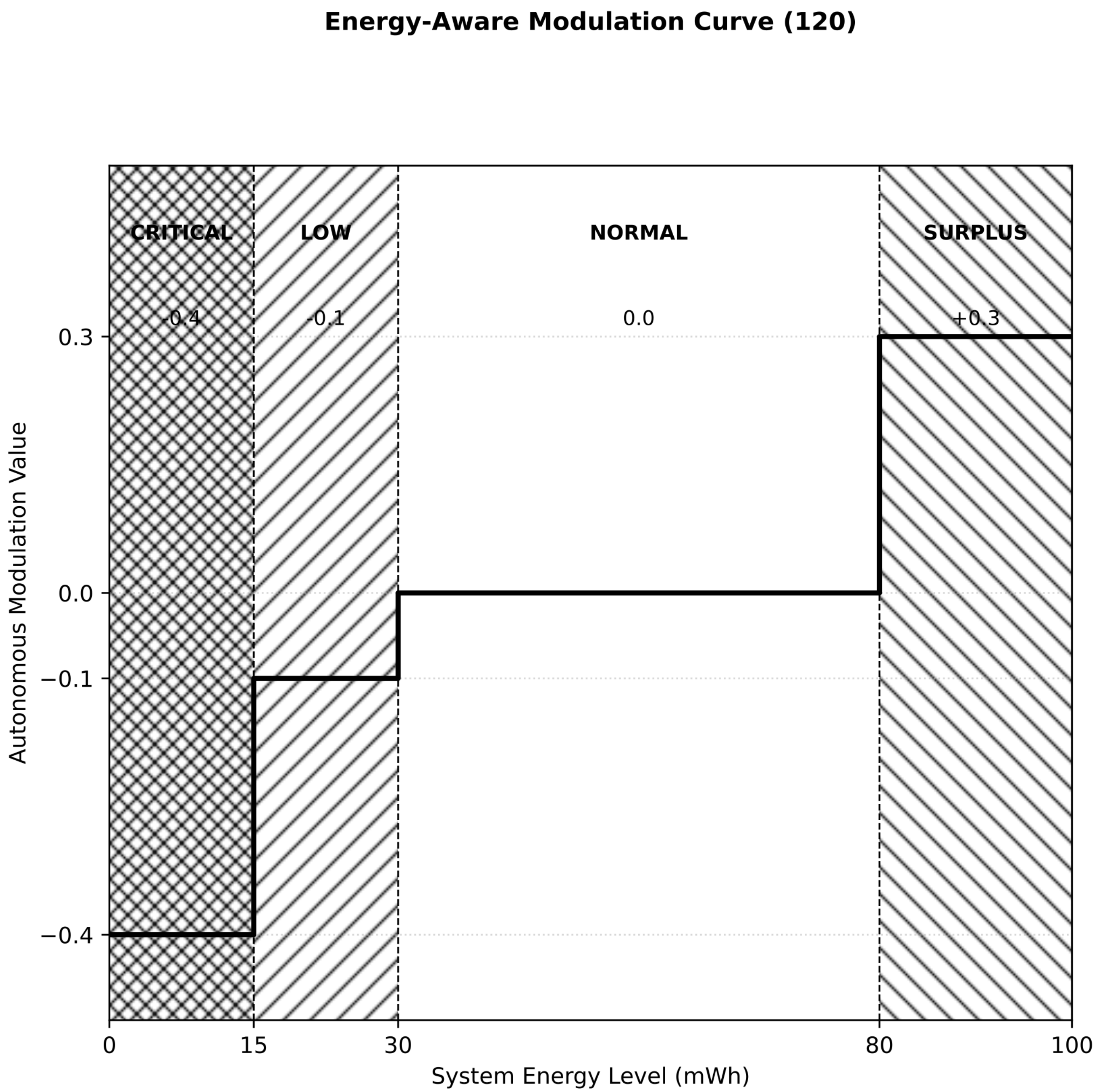

State Transition Diagram
Start
Heartbeat timeout
(>30s no tool call)
CONNECTED
Cognitive layer active
External input
External modulation
AUTONOMOUS
Fallback active
Self-generated input
Energy-aware mod.
Each step
(up to 10,
Any tool call
(direct reconnect)
Snapshot every 50 steps
Resync complete
resync() called
RECOVERING
Resynchronization
Transmitting buffer
Hard cap: 10,000 autonomous steps
Standard transition
Direct reconnection

FIG. 2



The system autonomously adjusts processing intensity based on energy

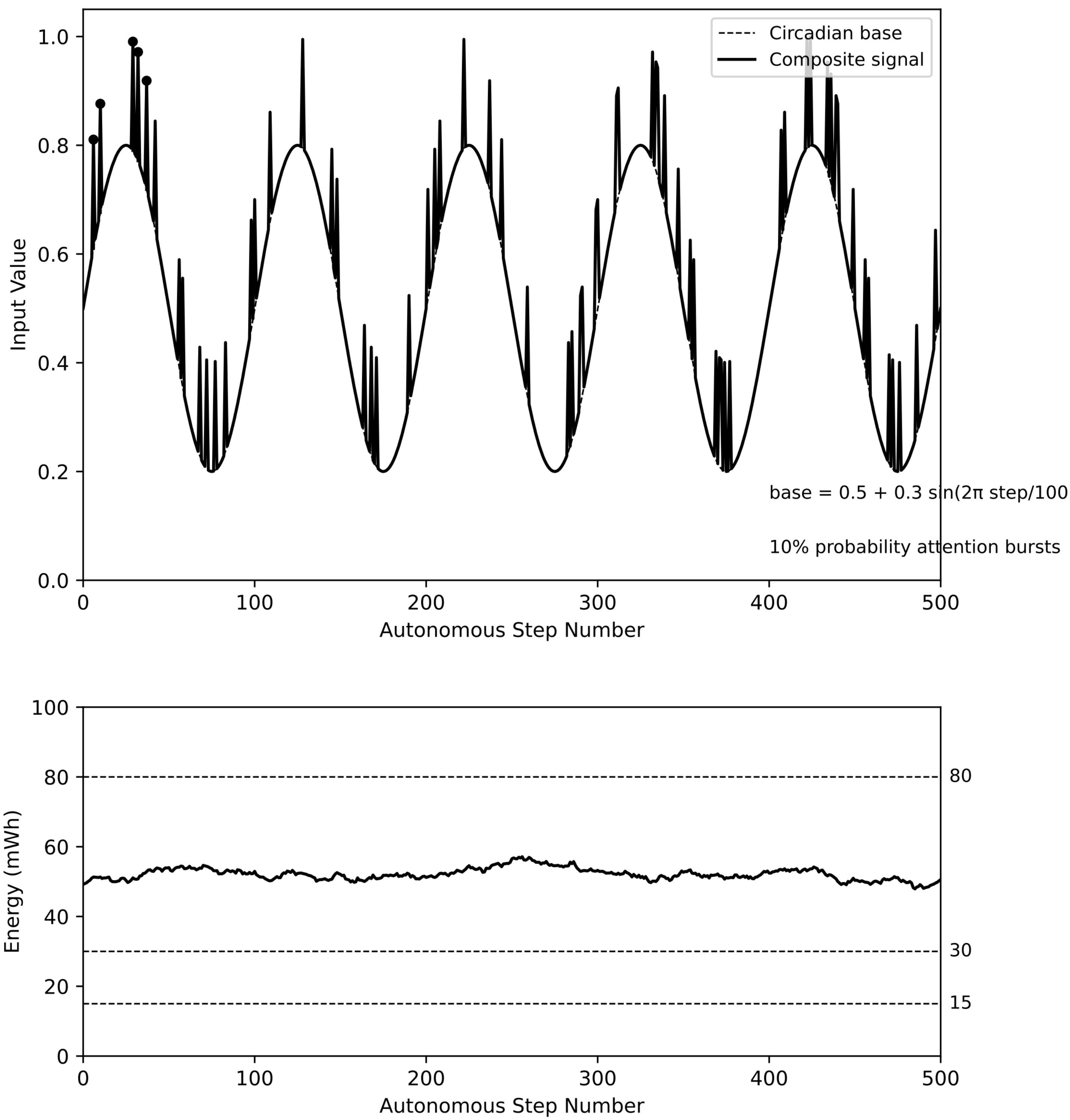
CRITICAL: Maximum conservation to prevent shutdown

SURPLUS: Opportunistic exploration using excess energy

Mimics biological metabolic regulation

FIG. 3

Autonomous Input Generator Output (118)



Circadian rhythm provides structured temporal variation

Energy level determines modulation via curve in FIG. 3

FIG. 4

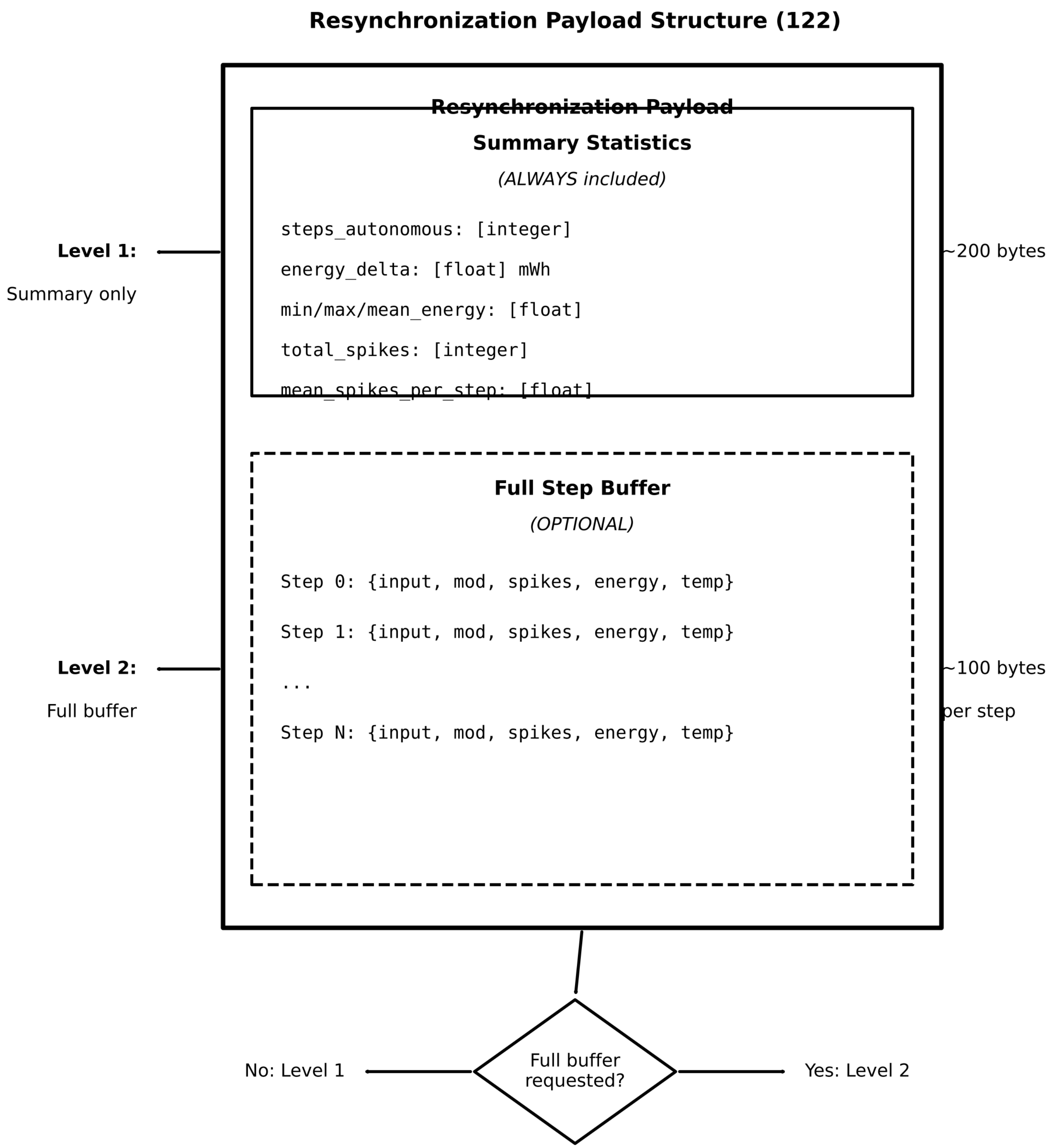

Resynchronization Payload Structure (122)
Resynchronization Payload
Summary Statistics
(ALWAYS included)
steps_autonomous: [integer]
energy_delta: [float] mWh
min/max/mean_energy: [float]
total_spikes: [integer]
mean_spikes_per_step: [float]
Level 1:
Summary only
~200 bytes
Full Step Buffer
(OPTIONAL)
Step 0: {input, mod, spikes, energy, temp}
Step 1: {input, mod, spikes, energy, temp}
...
Step N: {input, mod, spikes, energy, temp}
Level 2:
Full buffer
~100 bytes
per step
Full buffer requested?
No: Level 1
Yes: Level 2

FIG. 5

Recovery Timeline Diagrams

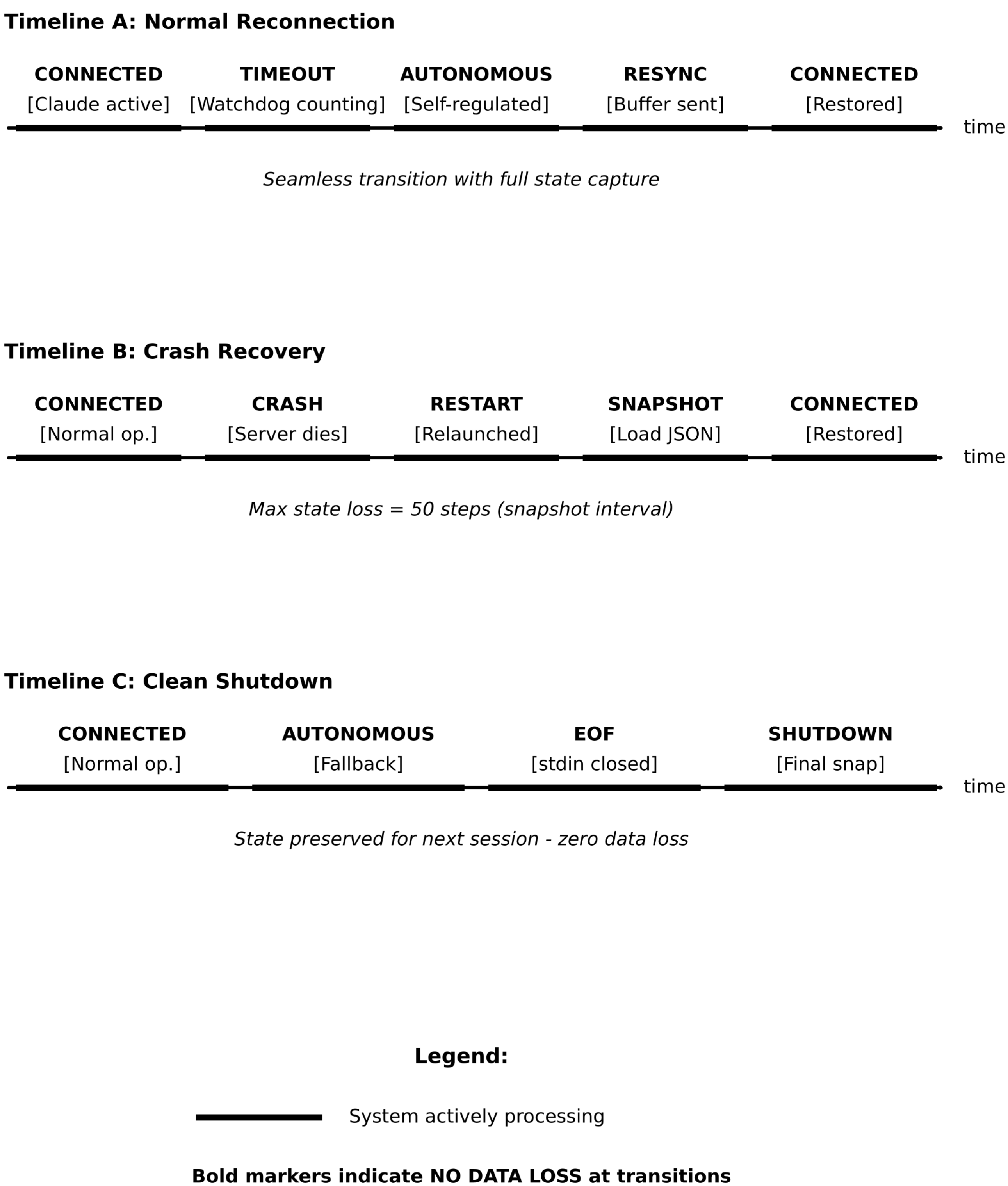
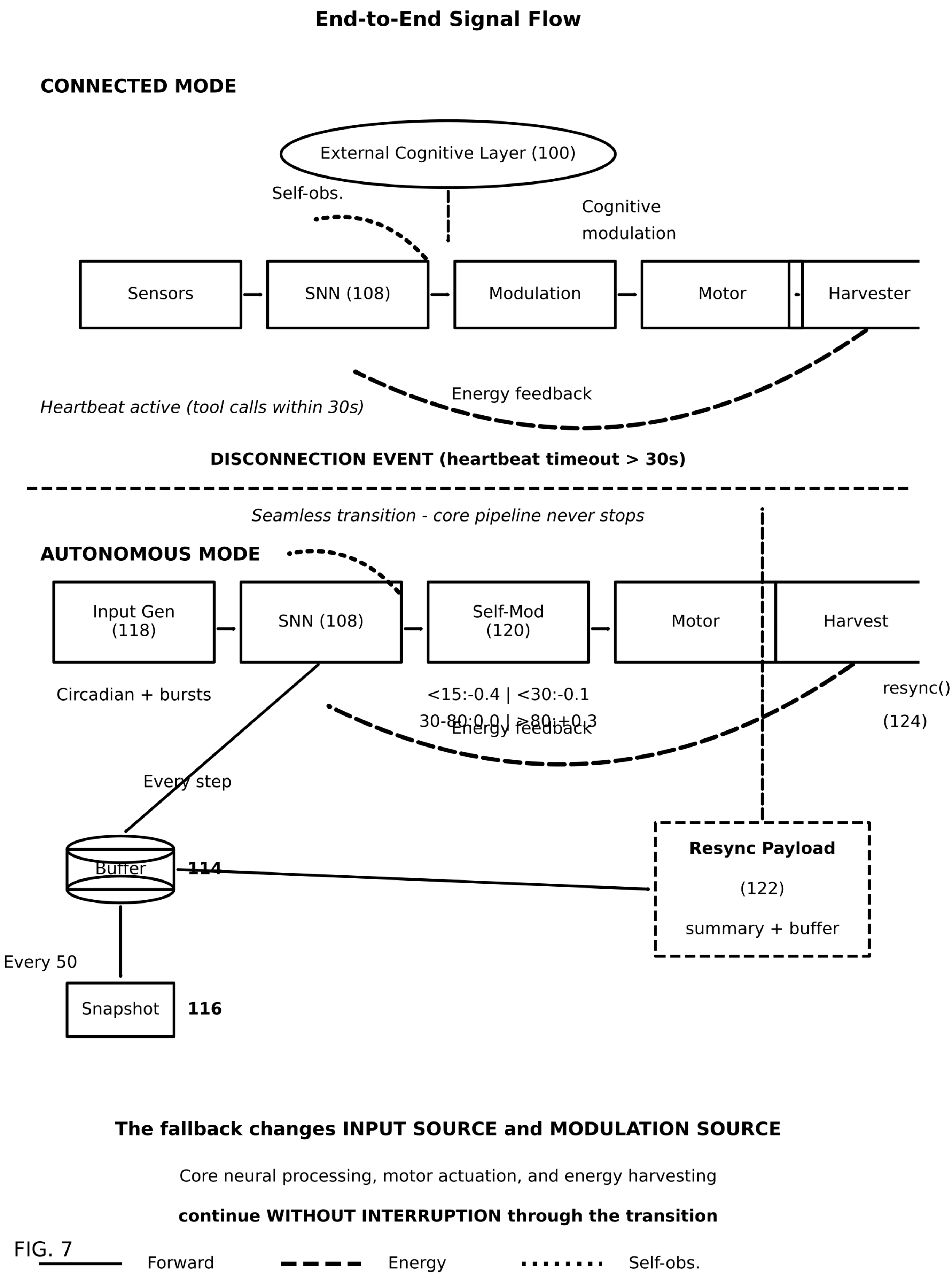


FIG. 6



End-to-End Signal Flow
CONNECTED MODE
External Cognitive Layer (100)
Self-obs.
Cognitive
modulation
Sensors
SNN (108)
Modulation
Motor
Harvester
Energy feedback
Heartbeat active (tool calls within 30s)
DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - core pipeline never stops
AUTONOMOUS MODE
Input Gen (118)
SNN (108)
Self-Mod (120)
Motor
Harvest
Circadian + bursts
<15:-0.4 | <30:-0.1
30-80:0.0 | >80:+0.3
Energy feedback
resync()
(124)
Every step
Buffer
114
Resync Payload
(122)
summary + buffer
Every 50
Snapshot
116
The fallback changes INPUT SOURCE and MODULATION SOURCE
Core neural processing, motor actuation, and energy harvesting
continue WITHOUT INTERRUPTION through the transition
Forward
Energy
Self-obs.

FIG. 7